Brief Communications

Severe Hypoglycemia During Maximized Insulin Treatment of Diabetes in a Randomized Clinical Trial

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Our experience with hypoglycemia during a randomized clinical trial testing the hypothesis of a causal relationship between hyperglycemia and diabetic complications is described. Patients under maximized control experienced hypoglycemia much more often than patients under standard control. There were five episodes of serious hypoglycemia, none of them on pumps, in the former group, but none in the latter. Lack of compliance (or overcompliance) with certain aspects of intensive treatment increases the danger of grave hypoglycemia and should be part of exclusion criteria when considering maximized control. DIABETES CARE 6: 62–63, JANUARY–FEBRUARY 1983.

The relationship between the prolonged hyperglycemia of diabetes mellitus and the associated microangiopathic complications remains controversial.[1-3] An increasing number of clinicians is now stressing improved blood glucose control. While the benefits of this approach are probable, although unproven, the risks have been poorly characterized and may remain largely unreported. We report here the serious consequences of hypoglycemia experienced by five diabetic patients, participants in the Minnesota Diabetes Complications Clinical Trial (MDCT), which tests in a randomized group of diabetic kidney recipients the hypothesis that "maximized" blood glucose control prevents or delays the development of diabetic nephropathy in renal allografts.[4] The patients were insulin-dependent with duration of disease of 13 yr or longer, metabolically unstable and on prednisone (as immunosuppression therapy), and all had been randomized to the maximized metabolic control group. Although approximately 40% of our patients in this group are treated with insulin pump, none of the patients described were on an insulin pump.

CASES

Case no. 1. D.J. was a 32-yr-old white male who joined our study 3 mo after the renal transplant. He was randomized to the maximized blood glucose control group, which includes intensive follow-up and aggressive insulin therapy. The daily treatment included two insulin injections (NPH and regular mixtures) and 3–4 home capillary blood glucose measurements. His blood glucose results were discussed by phone each afternoon and the insulin dose changed if needed. The blood glucose concentrations, not unlike those of several other unstable patients in the study, were within the range of 50–250 mg/dl. He was having an average of six hypoglycemic reactions per month and had several times failed to comply with advice aimed at improving his control and minimizing the danger of hypoglycemic reactions. He had refused to use an insulin pump. On July 13, 1981, D.J. was found unconscious at 2:30 p.m. He missed his mid-morning snack and lunch and slipped into hypoglycemic coma of possibly up to 5–6-h duration. His blood glucose concentration before breakfast had been 125 mg/dl. Upon arrival at the local hospital the blood glucose concentration was 12 mg/dl and he was unresponsive to deep pain. Intensive treatment failed to improve responsiveness and the patient remained in coma for 4 wk. Then he gradually improved, but has continued to show mild paranoid ideation, slightly inappropriate affect, and abnormal memory. His memory quotient (M.Q.) 6 mo after the comatose episode was 73 on the Weschler Memory Scale (normal: 90–110).

Case no. 2. M.P. was a 24-yr-old white male. He experienced symptoms of hypoglycemia at 4 p.m. one afternoon at which time his blood glucose measured 67 mg/dl. Failing to follow instructions, he chose to wait 1 h for the evening meal, developed convulsions, and fell and fractured the left femur. During surgery to repair the fracture the patient suffered a cardiac arrest and has remained in coma for the last 6 mo.

Case no. 3. J.M. was a 33-yr-old white male under the same maximized treatment who missed a mid-morning snack and had a severe hypoglycemic reaction while driving. He lost his job because of this accident.

Case no. 4. T.O. was a 26-yr-old white male, treated with three injections of regular insulin supplemented with NPH in the evening. During a hypoglycemic reaction while climbing stairs he fell down seven steps. Patient had resisted advice to decrease the insulin dose a few days before.

Case no. 5. C.C. was a 32-yr-old white female who persisted in having hard liquor during weekends. She had two hospital admissions for hypoglycemic coma of several days duration. She has recovered without sequelae.

The five cases were seen during the follow-up of more than 60 patient-yr of maximized treatment. During approximately 40 patient-yr of standard control (patients randomized to the standard treatment group) there were only three moderately severe hypoglycemic reactions leading to emergency room visits but no hospital admissions. Overall, the patients in the maximized group had hypoglycemic episodes approximately 10 times more often than those randomized to the standard group. At least six other patients under maximized treatment also have compliance problems but have not had serious hypoglycemia. The patients in the standard group took one (or in rare cases two) insulin injection daily and were followed much less intensively (once a month) than the patients in the maximized group. There was a large difference between the mean blood glucoses of the two groups (maximized group: 130 ± 9 mg/dl; standard group: 242 ± 22 mg/dl). Approximately 15% of the values are below 50 mg/dl in the intensive treatment and only 1% in the standard treatment.

DISCUSSION

These case reports—from our extensive experience with long-term, maximized treatment of diabetic patients—suggest that life-threatening hypoglycemic episodes, although unusual, are more often seen in patients under maximized diabetic control (in spite of very close follow-up) than in patients under standard control. All five patients reported were not compliant (or were overcompliant) with some of the guidelines provided to prevent hypoglycemia.

In addition to the patients reported here, we have had a few other patients in the maximized control group who have developed apparent tolerance to blood glucoses of 40 mg/dl or below with no symptoms. This tolerance, a complacent attitude of some of the patients toward symptomless hypoglycemia, and a marked compulsion to treat themselves aggressively when the blood glucose is only slightly elevated, has resulted in several overcompliant patients. This behavior of overcompliance or "sick role" has been seen in approximately 15% of our maximized control patients and poses serious management and ethical problems. Thus, both under- and overcompliance may pose problems regarding serious hypoglycemia in long-term maximized treatment of diabetic individuals.

We have modified several of the guidelines of our clinical trial to minimize the problems reported above. These modifications have included the relaxation of metabolic control in some of the more unstable patients, administration of insulin only at the time of the meal if the blood glucose is 70 or below, careful drilling of patients and their families in the management of severe hypoglycemia including the use of glucagon, and precautions to keep patients from being alone for any longer than 2 h.

A recent report[5] did not find hypoglycemia more common with intensive treatment. Our long-term studies of memory have not found any deterioration of function in maximized control patients,[6] except patient no. 1.

This report should not be misconstrued as a condemnation of maximized blood glucose treatment. Our goal is to alert the medical community to the unusual but potentially serious consequences of hypoglycemia in patients under vigorous insulin treatment, especially if these patients are metabolically unstable and/or noncompliant with the precautions aimed at preventing hypoglycemia.

In conclusion, certain under- and overcompliant patients may not be suitable for maximized insulin treatment, and a psychological evaluation to screen out these patients should be undertaken before initiating intensive treatment. This is particularly critical in the context of clinical trials, as it is in our case, since patients, once enrolled in the study, must be continued until the end.

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